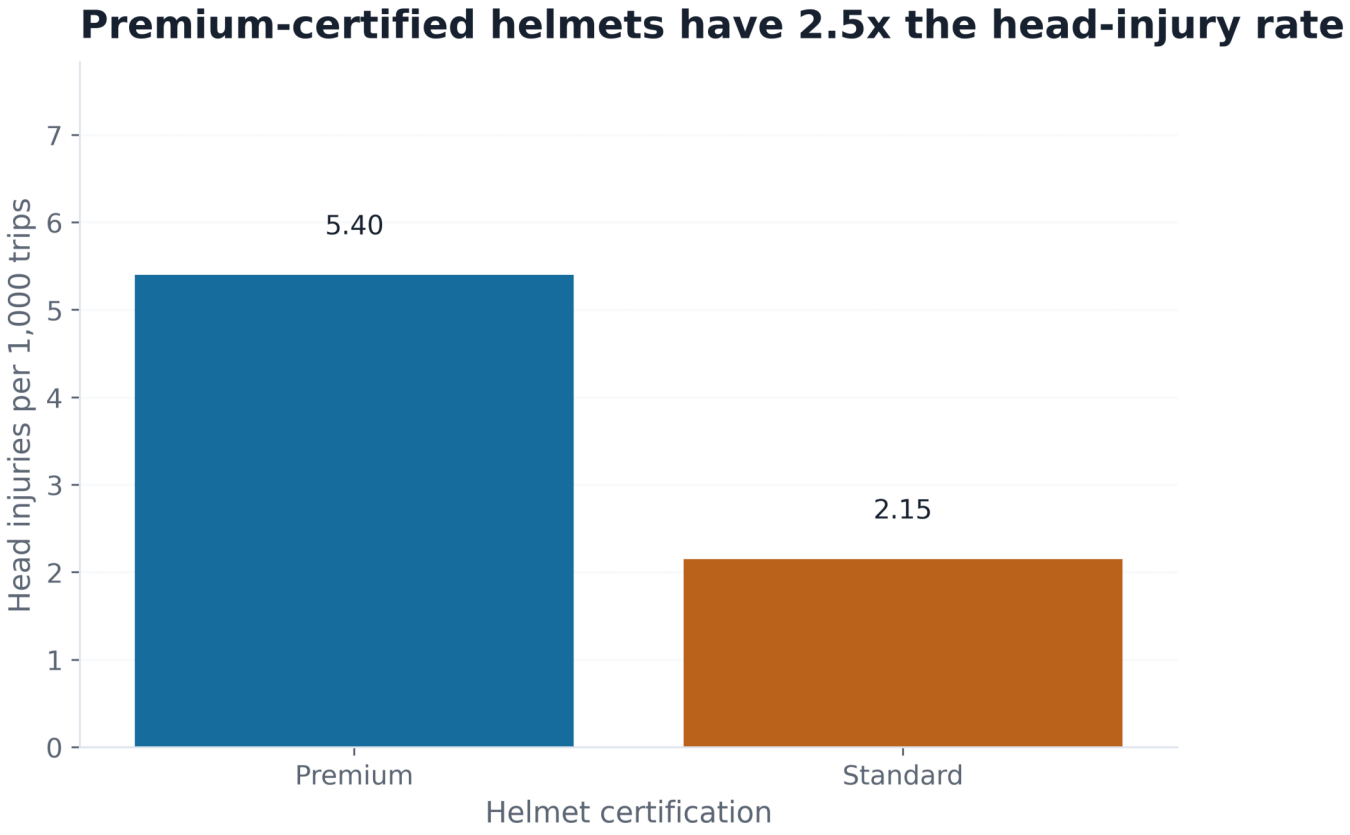
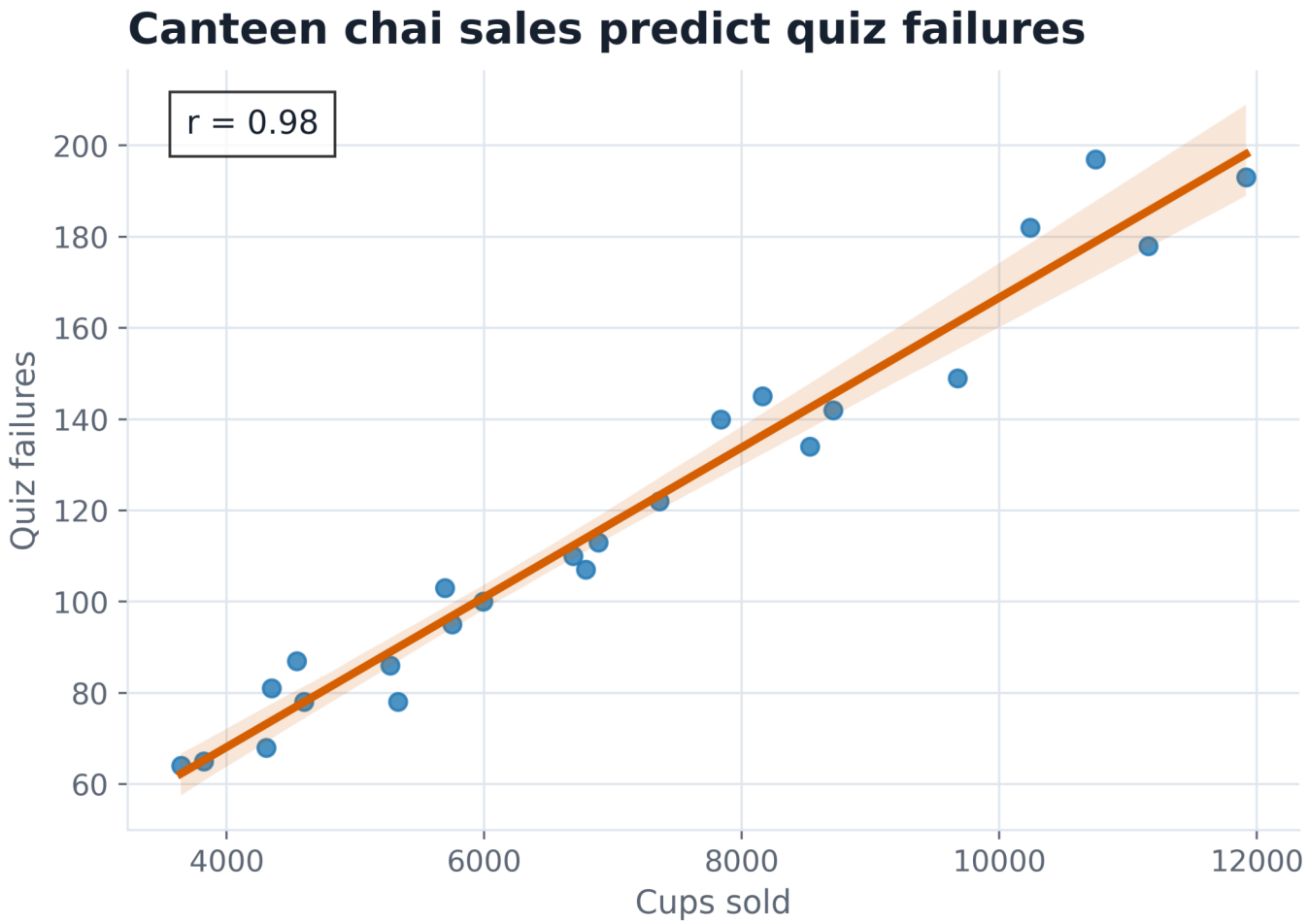
Case 1

The bar chart is reproduced using injuries per 1,000 trips; the real issue is the baseline and the hidden r



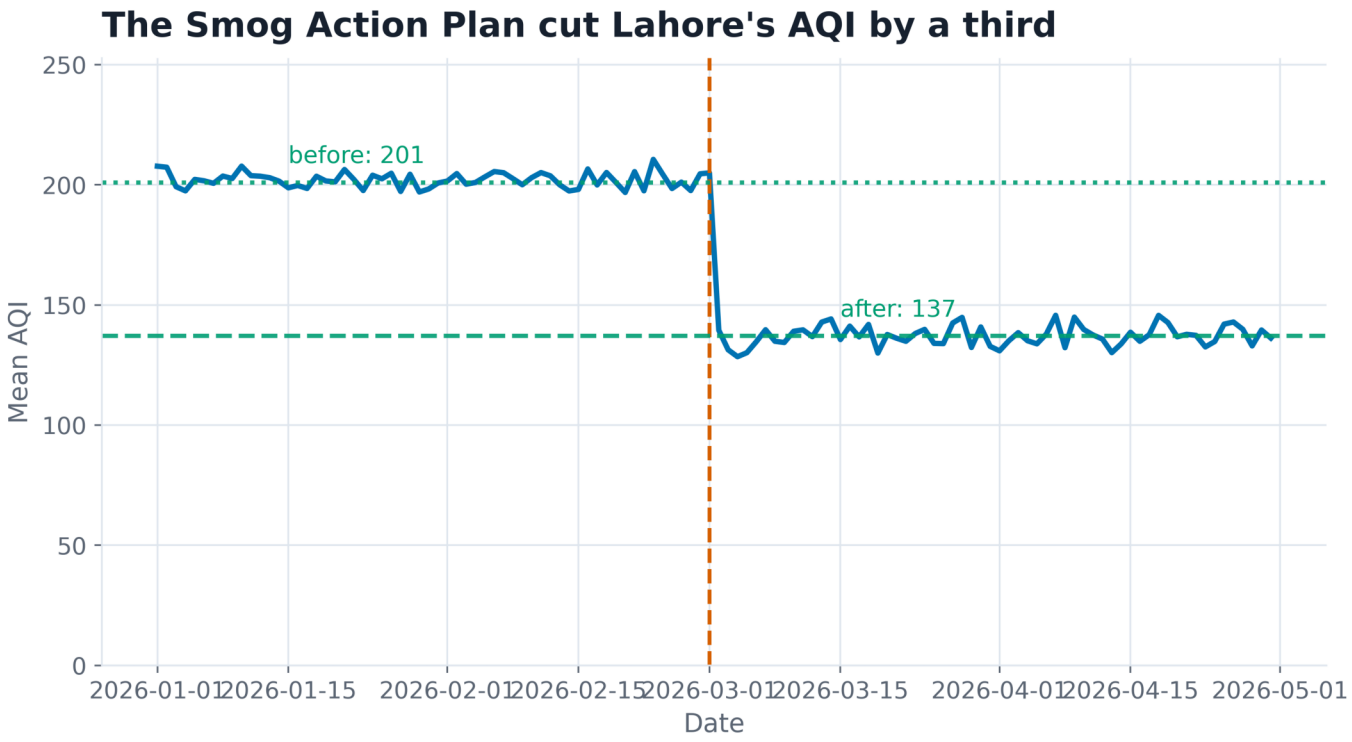
Case 2

The linear relation is real, but the mechanism is the monthly demand-and-exam cycle rather than a stab



Case 3

The apparent drop is partly a baseline choice: the chart compares mean across all stations after a policy



Case 4

The same summary statistics are not the same data distribution; the table hides non-linear structure an

Four departments behave identically in the summary table

department	n	mean hours	mean score	SD hours	SD score	r	slope
Architecture	11	18.0	14.5	6.63	2.03	0.817	0.250
Business Administration	11	18.0	14.5	6.63	2.03	0.816	0.250
Computer Science	11	18.0	14.5	6.63	2.03	0.816	0.250
Electrical Engineering	11	18.0	14.5	6.63	2.03	0.816	0.250

Case 5

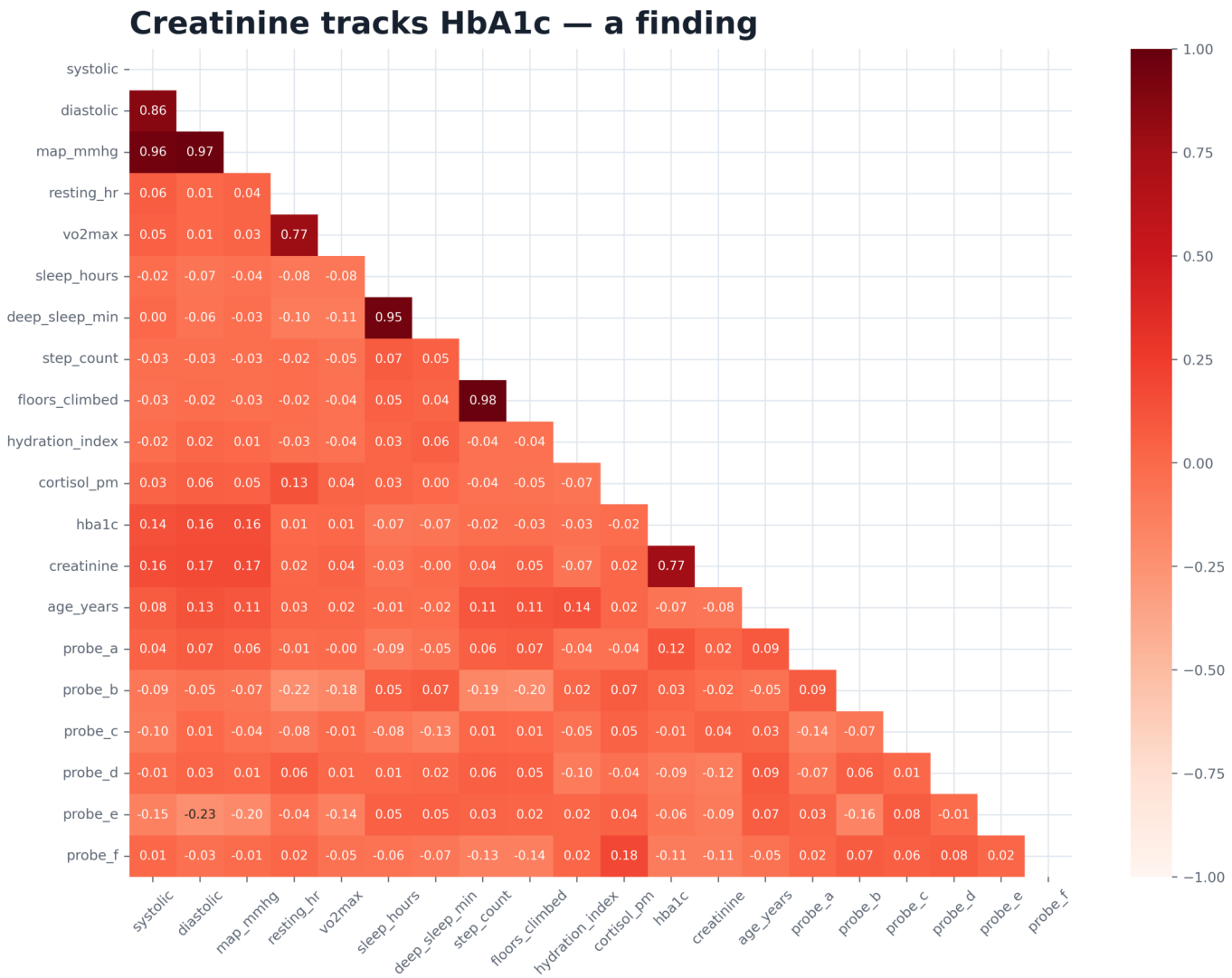
The cohorts are forced to identical moments, but the shape differences are masked in the table; the cha

Five cohorts share identical summary statistics

shape_name	n	mean h	mean score	SD h	SD score	r
ring	180	48.0	63.0	12.03	15.04	0.420
star	180	48.0	63.0	12.03	15.04	0.420
straight line	180	48.0	63.0	12.03	15.04	0.420
the letter X	180	48.0	63.0	12.03	15.04	0.420
two clusters	180	48.0	63.0	12.03	15.04	0.420

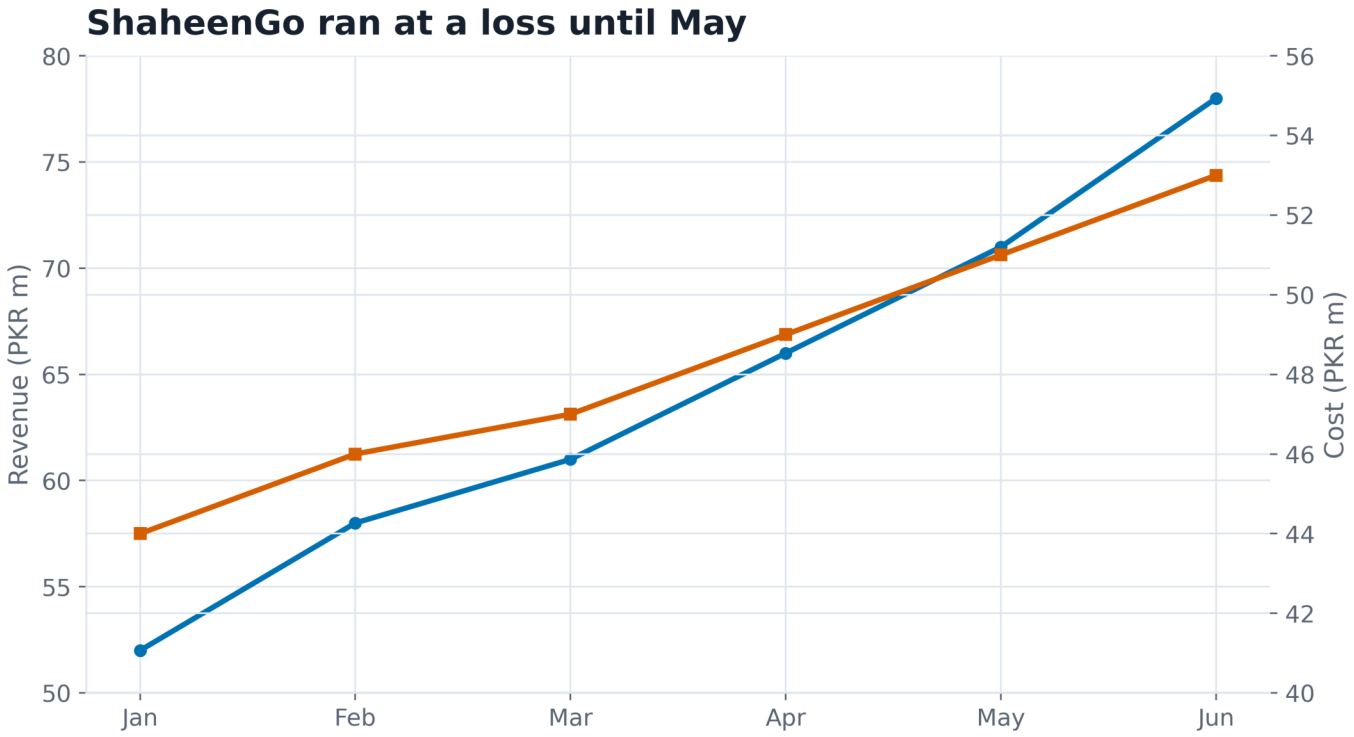
Case 6

The heatmap drawing issue is the unbounded red scale and the no-mask display; the real scientific issue



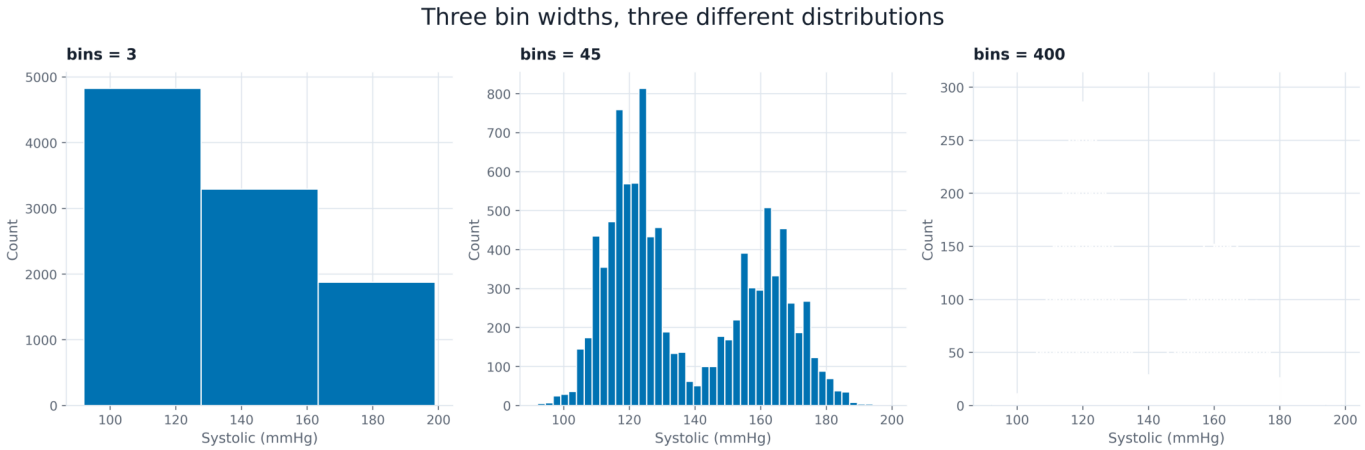
Case 7

The dual axis is a distortion: the cost and revenue series are shown on different scales, so the eye comp



Case 8

Binning choice changes the story; the data are the same, but bin width is a design choice and the distrib



Case 9

The KDE is not a distribution of observed values, and the plotted density extends beyond the physical ce

